

Deep Learning Fundamentals

Artificial Neural Networks



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Chapter 1

Artificial Neural Networks (ANN)

1.1 Introduction to Deep Learning & ANNs

1.1.1 Foundations and Background

- What is Deep Learning
- DeepLearning Vs Machine Learning
- Why deep learning getting famous
- Deep learning application
- deep learning types
- History and evolution of Deep Learning

1.2 The Perceptron

1.2.1 The Fundamental Building Block of Neural Networks

- What is Perceptron
- Perceptron vs sigmoided Neuron
- Limitations of single-layer perceptrons

1.3 Multi-Layer Perceptrons (MLP)

1.3.1 Architecture and Forward Propagation

- Internal architecture of MLPs
- Activation functions
- Forward propagation

1.3.2 Training Neural Networks

- Backpropagation
- Gradient Descent
- Optimization algorithms

1.4 Model Optimization and Tuning

1.4.1 Improving Network Performance

- Vanishing and exploding gradients
- Weight initialization
- Batch normalization
- Dropout
- Regularization
- Hyperparameter tuning

Chapter 2

Practical Implementation and Advanced Topics

2.1 Practical Setup with Keras & TensorFlow

- Installing TensorFlow and Keras
- Regression models
- Classification models
- Callbacks
- TensorBoard
- Functional API
- Saving and loading models

2.2 Training Performance Summary

Network Type	Epochs	Optimizer	Accuracy
Perceptron	10	SGD	0.750
MLP (Basic)	50	Adam	0.882
MLP (Regularized)	100	Adam	0.945

Table 2.1: Training Performance Summary

2.3 Key Equations in ANN

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial net} \cdot \frac{\partial net}{\partial w_j}$$